

Full Paper

Radiative Ignition of Fine-Ammonium Perchlorate Composite Propellants

Jeremy Cain, M. Quinn Brewster*

Department of Mechanical Science and Industrial Engineering, University of Illinois, Urbana, IL 61801 (U.S.A.)

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Abstract

Radiative ignition of quasi-homogeneous mixtures of ammonium perchlorate (AP) and hydroxyterminated polybutadiene (HTPB) binder has been investigated experimentally. Solid propellants consisting of fine AP (2 μm) and HTPB binder (~76/24% by mass) were ignited by CO₂ laser radiation. The lower boundary of a go/no-go ignition map (minimum ignition time vs. heat flux) was obtained. Opacity was varied by adding carbon black up to 1% by mass. Ignition times ranged from 0.78 s to 0.076 s for incident fluxes ranging from 60 W/cm² to 400 W/cm². It was found that AP and HTPB are sufficiently strongly absorbing of 10.6 μm CO₂ laser radiation (absorption coefficient $\approx 250 \text{ cm}^{-1}$) so that the addition of carbon black in amounts typical of catalysts or opacifying agents (up to 1%) would have only a small influence on radiative ignition times at 10.6 μm . A simple theoretical analysis indicated that the ignition time-flux data are consistent with in-depth absorption effects. Furthermore, this analysis showed that the assumption of surface absorption is not appropriate, even for this relatively opaque system. For broadband visible/near-infrared radiation, such as from burning metal/oxide particle systems, the effects of in-depth absorption would probably be even stronger.

Keywords: Radiation, Ignition, AP/HTPB Solid Propellant

1 Introduction

Thermal radiation plays an important role in ignition of many solid propellant systems. Radiation from burning aluminum (Al) droplets and molten aluminum oxide (Al₂O₃) smoke particles produced by the igniter to the main propellant grain is an important heating and ignition mechanism in large-scale solid rocket motors such as Space Shuttle and Ariane. While many studies have been conducted on radiant ignition of solid propellants [1–18], it is still not possible to predict or reliably model the ignition behavior of a new propellant system. Even for the simplest systems comprised of a homogeneous material subject to monochromatic radiation, the fundamental physics and chemistry are sufficiently complex that a comprehensive understanding and modeling capability is still lacking. The problem becomes much more difficult if one considers systems complicated by the issues of heterogeneity (e.g.

composite propellants) and/or broadband radiation. Nevertheless, progress has been made (as noted below) for both homogeneous (e.g. double-base) propellants [2–5, 9, 11, 15, 17] and composite (e.g. AP) propellants [2–4, 10, 11, 16–19] with both monochromatic [1–6, 8–10, 11–14] and broadband [2, 3, 7, 13, 15–18] radiation.

In this work, radiant ignition was studied for an important AP-composite propellant system that has not been reported previously: a mixture of very fine (2 μm) ammonium perchlorate (AP) and hydroxyterminated polybutadiene (HTPB) binder using monochromatic (10.6 μm) radiation. This system has attracted interest in recent years in the form of bi-modal propellants with wide AP size distribution (2 and 200 μm) [20] for its unique burning rate properties. The mixture of fine-AP and binder forms a quasi-homogeneous matrix within which the coarse AP particles sit. This mixture is fuel-rich because the fine AP is relatively uniform in size, even for close packing, with an upper limit of about 76% AP. Any particulate additives that might be included in the propellant, such as metal fuel (e.g. Al), burning-rate catalyst (e.g. Fe₂O₃), and opacifier (e.g. carbon black), reside in this fuel-rich matrix. Since this matrix region of the propellant between coarse AP particles (sometimes referred to as pocket propellant) is the most opaque region to broadband radiation produced by burning Al/Al₂O₃ (coarse AP is relatively transparent to such radiation), it is important to understand its radiant ignition behavior.

2 Previous Radiant Ignition Experimental Studies

One of the earliest reported investigations of radiative ignition and extinction of energetic solids was conducted by Ohlemiller et al. [5], in which double-base propellant was ignited by broadband radiation from an arc-image. Their observations showed that there is a minimum exposure time for a given radiant heat flux (or minimum flux for a given exposure time) below which self-sustained burning does not occur. In addition to this expected lower ignition limit, they discovered an upper limit that depended on pressure (Figure 1). Subsequent studies by De Luca et al. [2, 3] showed that the lower limit also depends on pressure, as well

* Corresponding author; e-mail: brewster@uiuc.edu

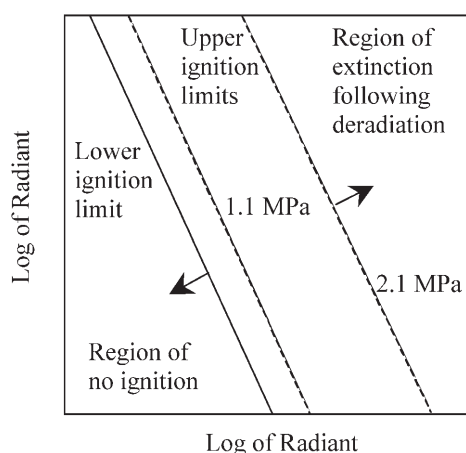


Figure 1. Radiative ignition map of Ohlemiller et al. [5] for a double-base propellant ignited by an arc-image furnace.

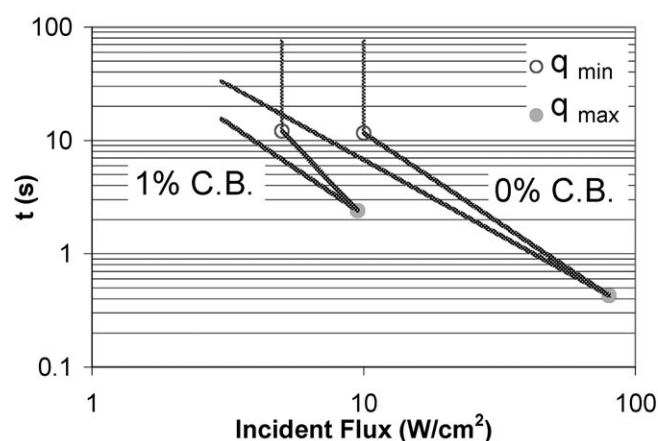


Figure 2. Radiative ignition map of Mikheev and Levashov [15] for a double-base propellant with a graphite source showing $q_{\max}$, $q_{\min}$ and effect of absorption coefficient.

as catalyst and opacifying agent concentration. For combinations of flux and heating time that were below the lower limit, the energetic solid simply had not absorbed enough heat to sustain burning when the radiation was removed (the surface temperature was still below the activation temperature for surface decomposition reactions). For flux-times above the upper limit, the combustion was over-driven by the radiation; the surface pre-heat zone was significantly thinner than that corresponding to non-irradiated burning, such that upon removal of the radiant flux, the surface layer burned off without establishing a sufficiently deep and hot preheated layer in the solid to support self-sustained burning.

At about the same time of the Ohlemiller et al. study, Mikheev and Levashov [15] reported similar findings for radiative ignition of a double-base propellant (Figure 2) using a graphite radiator (broadband source). Their results brought to light additional significant features of radiant ignition behavior that were not depicted in the findings of

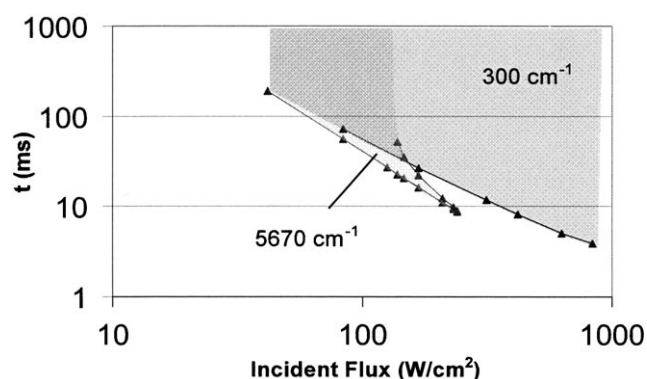


Figure 3. Numerical radiant ignition map of Weber et al. [1] using ZN theory and simplified kinetics for HMX, showing same features as Figure 2: $q_{\max}$, $q_{\min}$ (see Figure 2) and effect of absorption coefficient.

Ohlemiller et al. They showed that the upper and lower limits converged to define an upper-limit flux ($q_{\max}$) and that the upper ignition boundary asymptotically approached a lower-limit flux ($q_{\min}$). For fluxes below $q_{\min}$, there is no upper limit on the ignition corridor. Near the flux marked $q_{\max}$, the region of self-sustaining deflagration becomes vanishingly small. In addition, Mikheev and Levashov also showed that increasing the opacity by adding carbon black shifted the ignition boundaries to lower flux-times. These features of the radiation ignition map (e.g. $q_{\max}$, $q_{\min}$, and effect of absorption coefficient) were eventually simulated computationally using Zeldovich-Novozhilov (ZN) theory with a simplified chemical kinetics combustion model by Weber et al. [1], as shown in Figure 3 for HMX. More detailed combustion models with complex chemistry and transport have also been developed for simulating radiant ignition of energetic solids [9, 21, 22]; however, computationally generated go/no-go ignition results like Figure 3 based on such models have not yet been reported.

3 Optical Properties of Fine-AP/HTPB

Radiant ignition of solid propellants is strongly influenced by the optical properties of its ingredients. Thus, the spectral distribution of incident radiation and the nature of the various ingredients are crucial in determining how the radiation propagates into the solid and where it is absorbed. Broadband radiation that is spectrally distributed in the visible and near infrared regions can be absorbed very differently than longer infrared wavelengths (e.g. CO_2 laser), depending on the spectral location of absorption peaks as determined by lattice vibration frequencies.

Infrared optical properties for AP and HTPB obtained by FTIR and a laser-spatial filtering technique are reported by Isbell et al. [23, 24]. These results show that near $10.6 \mu\text{m}$, AP has an absorption peak corresponding to the perchlorate ion breathing frequency ($10.63 \mu\text{m}$) that is sharp but relatively weak compared with other lattice vibration absorption peaks; nevertheless, it has a strong influence on

Table 1. Composition of opacified fine-AP/HTPB propellants.

$f_{v,CB}$ (%)	K_a (cm ⁻¹)	$f_{m,AP}$ (%)	$f_{m,HTPB}$ (%)	$f_{m,CB}$ (%)	$f_{v,AP}$ (%)	$f_{v,HTPB}$ (%)
0.00	253.36	78.35	21.65	0.00	63.32	36.68
0.23	259.56	78.15	21.61	0.25	63.16	36.61
0.71	272.64	77.73	21.51	0.79	62.83	36.46
1.24	286.75	77.28	21.41	1.37	62.47	36.29

Table 2. Composition of non-opacified fine-AP/HTPB propellant.

Abbreviated Name	Chemical Name	Chemical Formula	Purpose	Weight %
R45M	hydroxyterminated polybutadiene	HO(C ₄ H ₆) _n OH	fuel (pre-polymer (binder))	17.92
IPDI	isophorone diisocyanate	OCNC ₆ H ₇ (CH ₃) ₃ CH ₂ NCO	curing agent	1.35
TEPANOL	tetraethylenepentamine acrylonitrile glycidol	(NH ₂ CH ₂ CH ₂ NHCH ₂ CH ₂) ₂ NHC ₂ H ₃ NC ₃ H ₆ O ₂	bonding agent	0.08
ODI	poly(tetrafluoroethylene oxide-co-difluoromethylene oxide) diisocyanate	CH ₃ C ₆ H ₃ (NCO)NHCO ₂ (CF ₂ CF ₂ O) _x (CF ₂ O) _z CONHC ₆ H ₃ (NCO)CH ₃	caking agent	0.03
DOA	dioctyl adipate	C ₂₂ H ₄₂ O ₄	plasticizer	4.65
AP	ammonium perchlorate (2 μm)	NH ₄ ClO ₄	oxidizer	75.97

the absorption coefficient near 10.6 μm because of its close spectral location. On the other hand, HTPB absorption near 10.6 μm is fairly flat, with influence from bond stretching occurring fairly far (spectrally) from 10.6 μm (for a detailed discussion of infrared absorption peaks, see Isbell et al. [23, 24]). AP and HTPB have reported absorption coefficients of 191 cm⁻¹ and 317 cm⁻¹, respectively, at 10.6 μm. Given the strong spectral content of the infrared absorption of these two materials, these values are remarkably similar at this particular wavelength, making the overall absorption of AP/HTPB relatively homogeneous. The absorption coefficient can also be interpreted as a photon mean free path, if scattering is neglected, as justified below. The photon mean free paths are approximately 50 μm and 30 μm, respectively, for AP and HTPB.

Not only is the absorptive power of AP and HTPB surprisingly similar at 10.6 μm, so is the refractive power. Isbell et al. [23, 24] report refractive index values of 1.75 and 1.73 for AP and HTPB, respectively. Since Fresnel interface reflection and scattering increase with increasing difference in refractive indices, the fact that these values are so close indicates that scattering of 10.6 μm radiation is minimal. To check this quantitatively, the Fresnel reflectance at the AP/HTPB interface can be estimated as $(1.75 - 1.73)^2 / (1.75 + 1.73)^2 = 0.000033$. Therefore, scattering of CO₂ laser radiation at the AP/HTPB interfaces is negligible and the AP/HTPB mixture can be treated as a purely absorbing, non-scattering, non-emitting medium.

An additional consideration in determining the absorption behavior of an AP/binder mixture is the opacity of the AP particles, which depends on their size and intrinsic absorption coefficient. AP particles that are nominally 2 μm in diameter with a bulk mean free absorption path of 50 μm are optically thin, and thus absorb volumetrically. The fine-AP/HTPB mixture absorption coefficient is therefore given by the volume-weighted average of the intrinsic absorption coefficients of the AP and HTPB binder.

The optical properties of samples used in this work were determined as follows. The AP/HTPB mixture used had a mass ratio of 76/24. With AP and binder densities of 1.95 and 0.93, respectively, the volume ratio is 63/37. These volume fractions and their intrinsic absorption coefficients (191 cm⁻¹ and 317 cm⁻¹) give a mixture absorption coefficient of 253 cm⁻¹. Carbon black was added in small amounts (<2% by mass) to increase the mixture's opacity, while not allowing the overall equivalence ratio to be changed significantly. The contribution to absorption from carbon black was calculated using classical Rayleigh scattering theory and optical constants for soot [25]. Table 1 shows the compositions, mass fraction (f_m) and volume fraction (f_v), and calculated absorption coefficients for the four mixtures studied. As indicated in Table 1, because AP and HTPB are fairly absorptive at this wavelength, the addition of small amounts of carbon black appears to have a relatively small effect on the overall absorption coefficient. Nevertheless, experimental testing was still needed to determine whether this small change in estimated absorption coefficient K_a causes a noticeable effect on the ignition behavior. The detailed composition of the AP/binder mix is given in Table 2.

Since a fraction of the radiation incident on the propellant surface is reflected at the surface and not available for heating the propellant, it is important to estimate this fraction. The reflectance at 10.6 μm for AP, HTPB, or their mixtures in air can be estimated by Fresnel theory as $(1.74 - 1)^2 / (1.74 + 1)^2 = 0.07$. The absorptivity is thus 0.93. This factor can be applied to the incident flux to estimate the amount of incident radiation absorbed.

4 Laser Ignition Tests

The experimental method was adapted from Weber et al. [1]. A multi-mode (quasi-Gaussian) laser beam of wave-

length $10.6\ \mu\text{m}$ was generated by a 1,000 W CO_2 laser, which had its temporal output characteristics (e.g. power output, pulse length, etc.) controlled by a function generator. The beam was expanded by two zinc-selenide lenses and sent to a beam integrator, where it was transformed into a spatially uniform profile and focused to a $1\ \text{cm}^2$ area at the sample location inside the combustion chamber. The sample was held in place on a stand by vacuum grease, and the chamber was open to atmospheric pressure but exhausted to remove combustion products.

Ignition tests were conducted according to the “go/no-go” criteria. This means that a laser pulse of constant flux level was applied over a given time duration. If the sample ignited and burned completely, it was counted as a successful ignition (“go”). If it did not ignite and burn out completely, it was counted as a “no-go.” For a given flux level, the pulse duration was varied to determine the time at which ignition occurred, as evidenced by self-sustained burning upon removal of radiant energy. Each test was recorded by a digital video camera. Attached to the chamber was an LED, which was viewed with the propellant by the camera during tests. A timing circuit caused the LED to emit visible light during the laser pulse. Video images of the tests were analyzed using video editing software. This made it possible to see when irradiation occurred via LED emission. This arrangement also made it possible to record the time at which light was first emitted from the propellant surface, the so-called “first-light” time.

5 Radiant Ignition Results

The radiant ignition results for all four propellants are shown in Figure 4. These data correspond to the lower ignition boundary indicated in Figures 1–3. The observed minimum ignition time varied from 0.78 s to 0.076 s for incident fluxes ranging from $60\ \text{W}/\text{cm}^2$ to $400\ \text{W}/\text{cm}^2$. One important feature of Figure 4 is the slope of the data. The

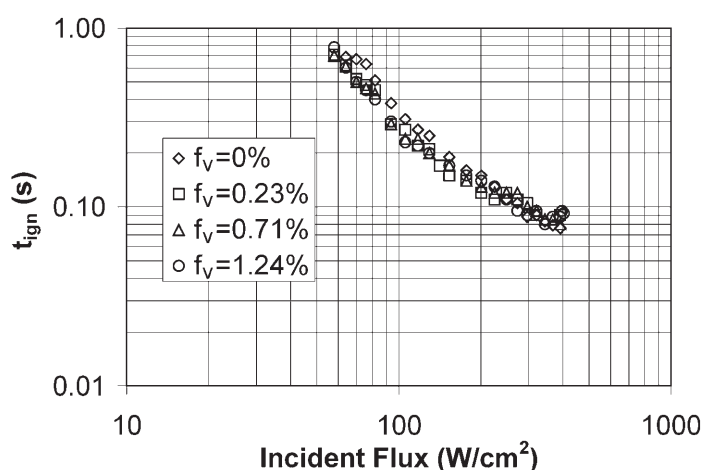


Figure 4. Experimentally determined radiant ignition lower boundary for optically modified fine-AP/HTPB propellants at $10.6\ \mu\text{m}$.

classical reference slope of -2 on a log-log plot of ignition time versus flux corresponds to surface absorption and a fixed surface-temperature (T_s) ignition criterion, as given by simple heat transfer theory. An analytic solution for a semi-infinite solid with a surface heat-flux boundary condition is found in most heat transfer texts [26]. The slope of the experimental data for no carbon black in Figure 4 varies between -1.7 and -1.0 , significantly less in magnitude than -2 . This is possibly indicative of in-depth absorption effects or breakdown of the surface temperature ignition criterion with increasing flux, as discussed further below.

Another feature to note is the effect of varying opacity (varying carbon black loading). The effect is rather minimal for these propellants at this wavelength; in fact it is barely noticeable above the experimental variability of $\pm 5\ \text{ms}$. This finding confirms the implication of the calculated optical properties noted above, that adding opacifying agent or catalyst particles at the level typical of these ingredients ($< 1\%$) has only a small effect on radiant ignition at $10.6\ \mu\text{m}$. Of course, radiation with different spectral distributions (e.g., shorter wavelengths) can demonstrate a stronger effect of the carbon black on opacity and ignition [2, 3].

6 Surface Absorption Assumption

In models of ignition of solid propellants where a radiant flux is present, the assumption of surface absorption (e.g. a surface heat flux boundary condition) is commonly invoked [9, 27–29]. A simple theoretical model can be developed to test the applicability of this assumption based on the 1-D heat conduction equation with in-depth volumetric absorption [30]. A collimated radiative flux is imposed on a semi-infinite, homogeneous solid. The propellant is characterized by an absorption coefficient (K_a), thermal conductivity (k), thermal diffusivity (α), and surface reflectivity (ρ). Ignition is assumed to occur at a fixed surface temperature, T_s . Thermodynamic property values over a broad range of temperatures are available for AP and HTPB [31–33]. Their values were averaged over the temperature range (initial to ignition temperature), and are $k = 0.251\ \text{W}/\text{m} \cdot \text{K}$ and $\alpha = 9.57 \times 10^{-8}\ \text{m}^2/\text{s}$. The absorption coefficient was taken as $250\ \text{cm}^{-1}$, corresponding to the unmodified AP/HTPB propellant. The ignition temperature was assumed to be $T_s = 925\ \text{K}$. The predicted results for ignition time with in-depth absorption are shown in Figure 5. For contrast, the case of surface absorption ($K_a \rightarrow \infty$) is also shown. The surface absorption case represents a limiting case with a slope of -2 corresponding to the analytical solution for a surface heat-flux boundary condition. For finite K_a , the minimum ignition time increases for a fixed flux, and more so at higher fluxes, resulting in a boundary slope magnitude less than 2. Thus, one of the possible causes for the shape of the boundary in Figure 4 is in-depth absorption. As Figure 5 shows, the model fit is not as good at higher fluxes as at lower fluxes. The simple ignition criterion of fixed T_s obviously does not capture the complex dynamics of radiant ignition, effects of which are discussed further in Weber et al. [1].

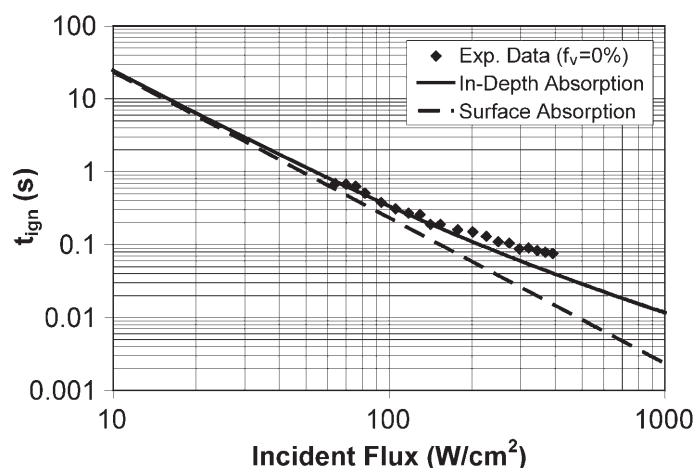


Figure 5. Radiant ignition lower boundary for optically unmodified fine-AP/HTPB propellants at $10.6\ \mu\text{m}$; experimental data with two theoretical curves: surface and in-depth absorption ($T_{s, \text{ign}} = 925\ \text{K}$, $K_a = 253\ \text{cm}^{-1}$).

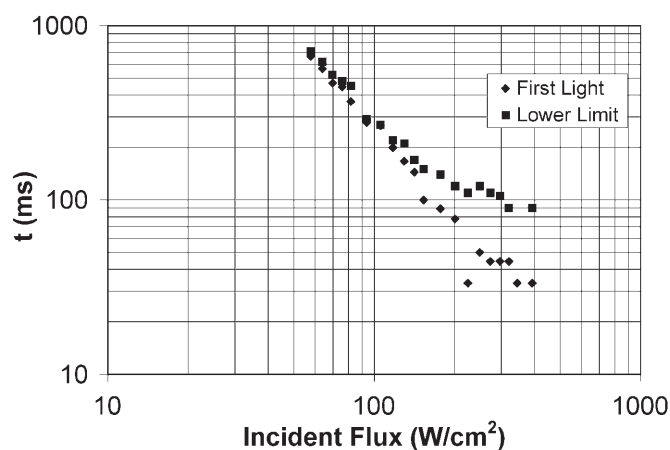


Figure 6. First-light and lower ignition boundary for $f_v = 0.23\%$.

7 First-Light Results

Figure 6 shows the first-light results for the composite with $f_v = 0.23\%$ soot. The results indicate that the ignition (go/no-go) time is significantly longer than the first-light time, with differences becoming more pronounced at higher fluxes. This suggests that first light, on its own, is not an accurate indication of ignition, particularly at high fluxes. Ignition tests at elevated pressures [2] suggest that the time between first light and the lower ignition boundary decreases with increasing pressure. The first-light time is also dependent on the wavelength of the light emitted.

8 Combustion Stability of Quasi-Homogeneous Fine-AP/HTPB

Quasi-homogeneous fuel-rich fine-AP/HTPB matrix propellants, such as that used here, are known not to be stable at

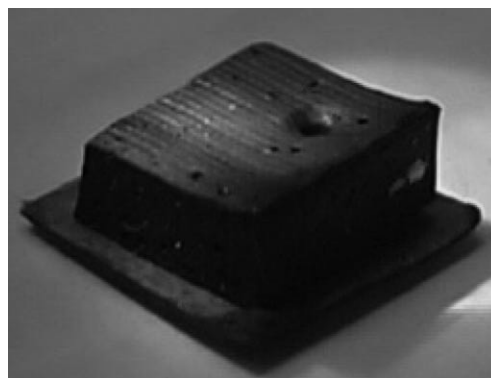


Figure 7. Sample that underwent self-extinguishment.

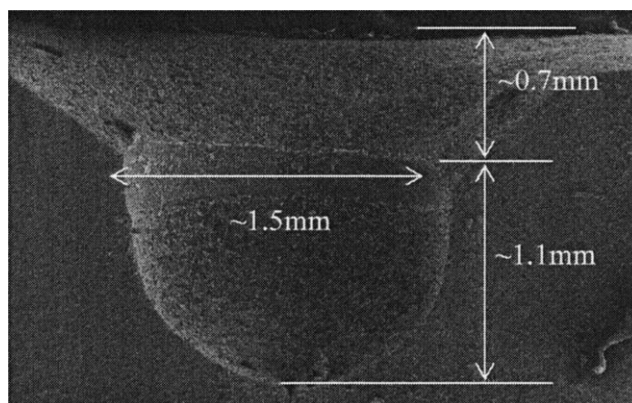


Figure 8. SEM photograph of regressed hole.

pressures above $0.1\ \text{MPa}$ [34, 35]. Under the $0.1\ \text{MPa}$ conditions of this study, the propellants burned stably upon ignition. However, conditions were found to be sufficiently close to the stability boundary associated with elevated pressure that occasionally a sample would self-extinguish. One such sample is shown in Figure 7. This type of instability exhibits non-planar burning with formation of pockets in the burning surface that grow and then extinguish themselves. Figure 7 shows the beginning of the development of such a pocket. To examine the condition of the extinguished surface, this sample was sectioned and SEM photographs taken. Figure 8 shows that this pocket has two distinct shapes: a conical top with a spherical region below.

A surprising result from the SEM investigation is shown in Figure 9. It was found that a narrow, long crack of uniform width had formed at the bottom of the hole. Figure 9 shows that it extended from the bottom of the hole through the entire sample. The crack is approximately $10\ \mu\text{m}$ wide and $3.2\ \text{mm}$ long, as shown by Figure 10. The mechanism by which such cracks might be created has not been discussed previously in the literature, nor has an observation of such a crack been reported, to the authors' knowledge. The fact that the crack originates at the bottom of the hole suggests that its creation had something to do with the hole's formation.

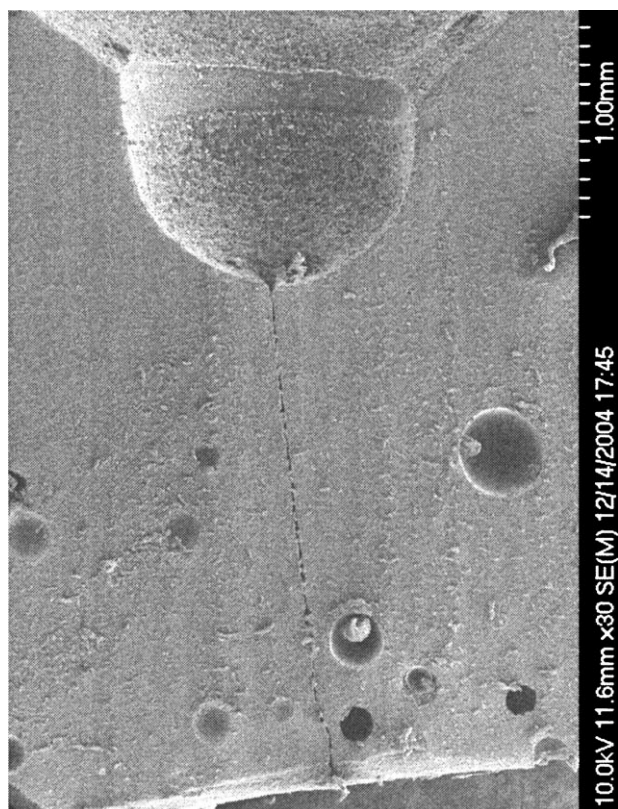


Figure 9. SEM photograph of crack.

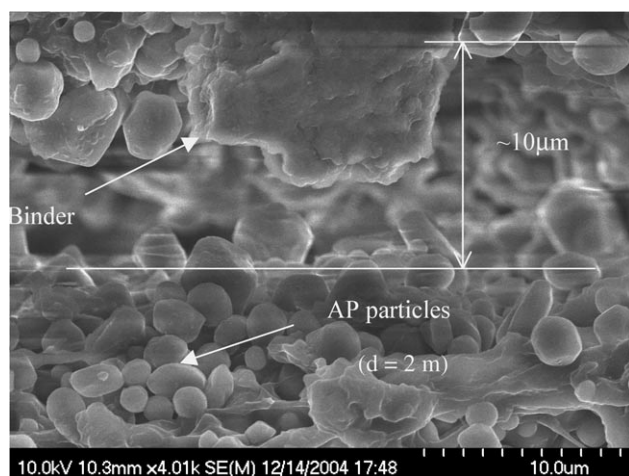


Figure 10. Magnified SEM photograph of crack.

9 Conclusions

Radiant ignition of fine-AP/HTPB propellant by CO₂ laser radiation is strongly influenced by the optical properties of AP and HTPB at 10.6 μm. In-depth absorption effects seem to be evident in the go/no-go ignition results, although other effects cannot be conclusively eliminated. It was found that AP and HTPB are sufficiently strongly absorbing of 10.6 μm CO₂ laser radiation so that the addition of soot in amounts typical of catalysts or opacity-modifying substances

(up to 2% by mass) have only a small influence on radiative ignition times (at this particular wavelength). A simple theoretical analysis indicates that the assumption of surface absorption is not appropriate, even for this relatively opaque system. For broadband visible/near-infrared radiation, such as from burning metal/oxide particle systems (i.e. Al/Al₂O₃), the effects of in-depth absorption would probably be even stronger due to even weaker absorption, as suggested by the optical property data of Isbell [23, 24].

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